What “In-Place” Means

In computer science, an in-
place algorithm rearranges or transforms its input using only a small, constant amount of extra memory. It operates directly on the original dat
[1] .

Strict vs. Broad Definitions

- Strict definition • Uses only Θ(1) extra space, counting everything (variables, pointers, recursion stack). • Extremely restrictive since even an i
- Broad definition • Allows small nonconstant extra space (often O(log n) for recursion or auxiliary variables). • Requires no extra space propo
  [1] .

Why In-Place Matters

- Reduces overall memory footprint, crucial on embedded or resource-limited systems
- Improves cache locality by operating within a single memory region
- Limits garbage collection or allocation overhead

Examples of In-Place vs. Out-of-Place Sorting

Algorithm	In-Place?	Extra Space
Bubble Sort	Yes	O(1)
Insertion Sort	Yes	O(1)
Selection Sort	Yes	O(1)
Heap Sort	Yes	O(1)
Quick Sort	Broadly yes	O(log n) recursion
Merge Sort	No	O(n) for merging
Timsort	No	O(n) temporary arrays

Looking ahead, understanding in-place constraints can inform design choices in streaming data algorithms, in-
place graph traversals, and low-memory neural network inference. Let me know if you’d like to dive into any of those!